Bridging Stereo Geometry and BEV Representation with Reliable Mutual Interaction for Semantic Scene Completion

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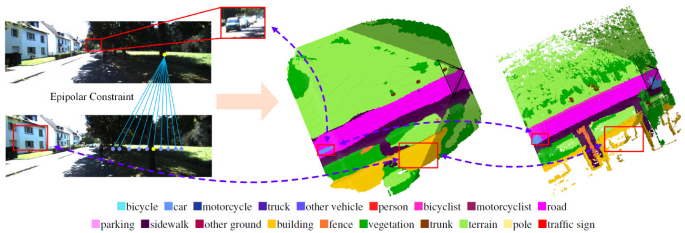
Motivations:

- 3D semantic scene completion (SSC) is an ill-posed perception task that requires inferring a dense 3D scene from limited observations.
- Previous methods struggle to predict accurate semantic scenes due to inherent geometric ambiguity and incomplete observations.
- In this paper, we resort to stereo matching technique and bird's-eye-view (BEV) representation learning to address such issues in SSC.

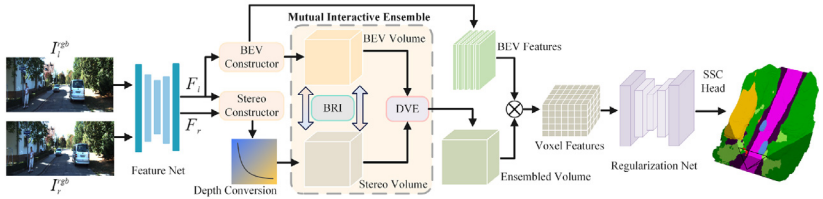


The figure illustrates the stereo inputs, SSC prediction results and ground truth from left to right. We can see that our method shows promising performance in completing semantic scenes, especially for those challenging distant small objects, as indicated by the car highlighted with a red box.

Contributions:

- A novel framework that resorts to both stereo and BEV representations with dense 3D volumes for precise geometry modeling and hallucination ability enhancement in SSC;
- A Mutual Interactive Ensemble block to bridge the representation gap and take advantage of the complementary merits of the volumes for fine-grained reliable perception;
- BRGScene outperforms state-of-the-art VoxFormer-T with a 14.5% relative improvement on the SemanticKITTI leaderboard.

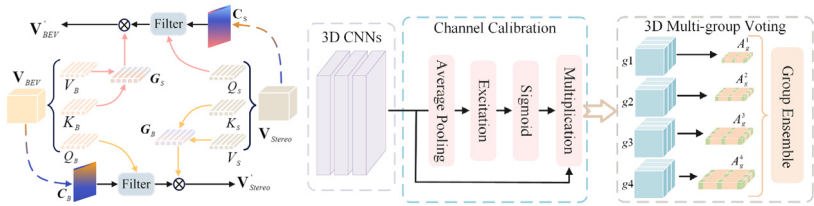
Bridging Stereo Geometry and BEV Representation(BRGScene):



- Our proposed BRGScene aims to jointly infer dense 3D geometry and semantics solely from RGB images.
- The BEV latent volume and stereo geometric volume are constructed by a BEV Constructor and a Stereo Constructor, respectively.

Bi-directional Reliable Interaction (BRI)

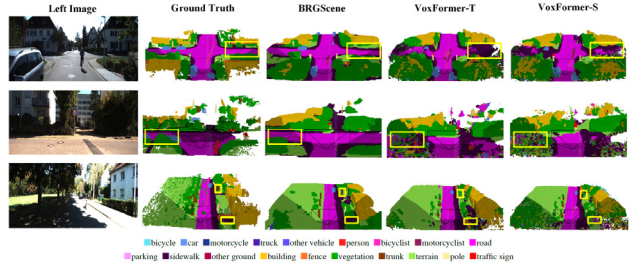
Dual Volume Ensemble (DVE)



- To bridge the representation gap for fine-grained reliable perception, a Mutual Interactive Ensemble block is designed to take advantage of the complementary merits of the volumes in the two representations, which includes Bi-directional Reliable Interaction (BRI) Module and Dual Volume Ensemble (DVE) module.
- The Bi-directional Reliable Interaction module is designed for pixel-level reliable geometry information interaction, while the Dual Volume Ensemble module is devised for mutually beneficial aggregation.

Experiments:

- Qualitative results on the SemanticKITTI validation set.



- Quantitative results on the SemanticKITTI test set.

Methods	BRGScene(ours)	VoxFormer-T(2023)	VoxFormer-S(2023)	TPVFormer(2023)	MonoScene(2022)	SSCNet(2017)
Input	Stereo	Stereo-T	Stereo	Mono	Mono	LiDAR
IoU (%) ↑	43.89	43.21	42.95	34.25	34.16	**49.98**
mIoU (%) ↑	**16.39**	13.41	12.20	11.26	11.08	16.14
car (3.92%)	**24.90**	21.70	20.80	19.20	18.80	24.26
bicycle (0.03%)	**2.60**	1.90	1.00	1.00	0.50	0.54
motorcycle (0.03%)	**1.80**	1.60	0.70	0.50	0.70	0.78
truck (0.16%)	**5.30**	3.60	3.50	3.70	3.30	1.18
other-veh. (0.20%)	**8.10**	4.10	3.70	2.30	4.40	4.43
person (0.07%)	**2.00**	1.60	1.40	1.10	1.00	0.25
bicyclist (0.07%)	2.30	1.10	**2.60**	2.40	1.40	0.25
motorcyclist (0.05%)	**0.90**	0.00	0.20	0.30	0.40	0.03
road (15.30%)	**62.10**	54.10	53.90	55.10	54.70	51.15
parking (1.12%)	**32.70**	25.10	21.10	27.40	24.80	27.12
sidewalk (11.13%)	**32.60**	26.90	25.30	27.20	27.10	30.76
other-grnd (0.56%)	**11.00**	7.30	5.60	6.50	5.70	6.44
building (14.10%)	26.00	23.50	19.80	14.80	14.40	**34.53**
fence (3.90%)	18.40	13.10	11.10	11.00	11.10	**19.87**
vegetation (39.3%)	24.90	24.40	22.40	13.90	14.90	**35.25**
trunk (0.51%)	**10.60**	8.10	7.50	3.70	3.30	1.18
terrain (9.17%)	28.00	24.20	21.30	20.40	19.50	**29.01**
pole (0.29%)	8.90	6.60	5.10	2.90	3.30	**13.10**
traf.-sign (0.08%)	**8.20**	5.70	4.90	1.50	2.10	6.73

BRGScene surpasses temporal stereo-based (Stereo-T) VoxFormer-T and LiDAR-based SSCNet in terms of mIoU.

Paper **Code**

Welcome to read more details!